Reducing overfitting in an LSTM resume classifier with AEDA

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What the notebook does

Step	What happens
Preprocessing	Clean the text, keep the first 400 words, split 70/15/15 by category (1,728 / 362 / 394 resumes), build a 15,000-word vocabulary from the training set only.
Dataset	Each resume becomes 400 word IDs (padded or cut). Labels are 24 job categories.
Model	Embedding, 2-layer LSTM, linear layer. The prediction uses the last hidden state.
Training	No dropout, no weight decay, Adam, 12 epochs. This is on purpose, so the model overfits.
Evaluation	Test accuracy, macro F1, loss curves, confusion matrix.
AEDA experiment	Same model trained on augmented data, then compared with the baseline.

The paper: AEDA

"An Easier Data Augmentation Technique for Text Classification. Karimi, Rossi and Prati, 2021 (arXiv 2108.13230)"

AEDA does one thing: it inserts random punctuation marks (. , ! ? ; :) into a text.

For a text with n words it adds between 1 and n/3 marks at random positions.

No word is removed or swapped, so the word order stays intact and the label stays the same. One resume turns into several slightly different training examples.

Original

python developer with five years of experience in machine learning

After AEDA

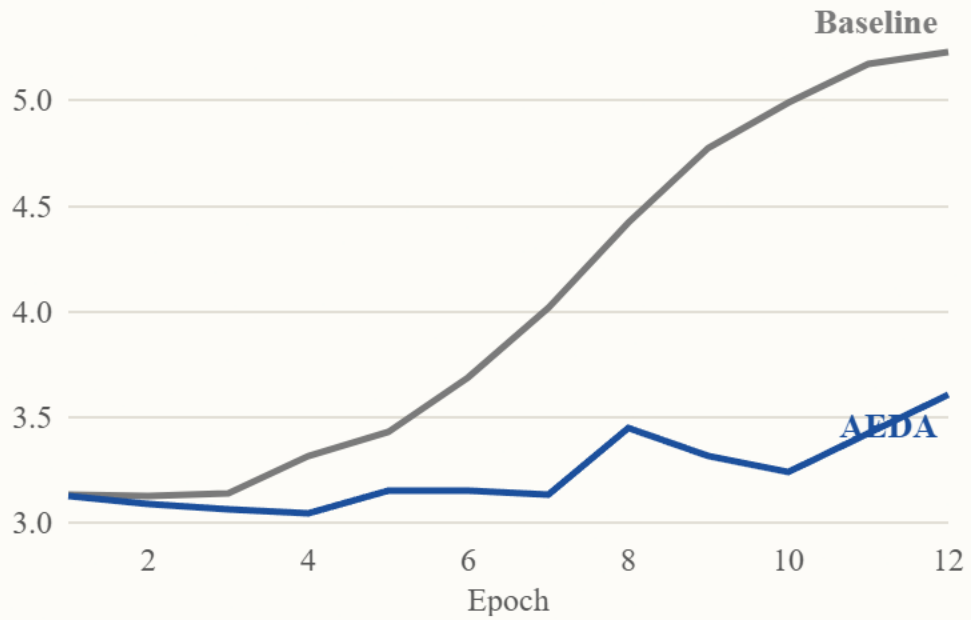
; python developer with five ? years of experience in machine ! learning

Same model, same settings, more training text

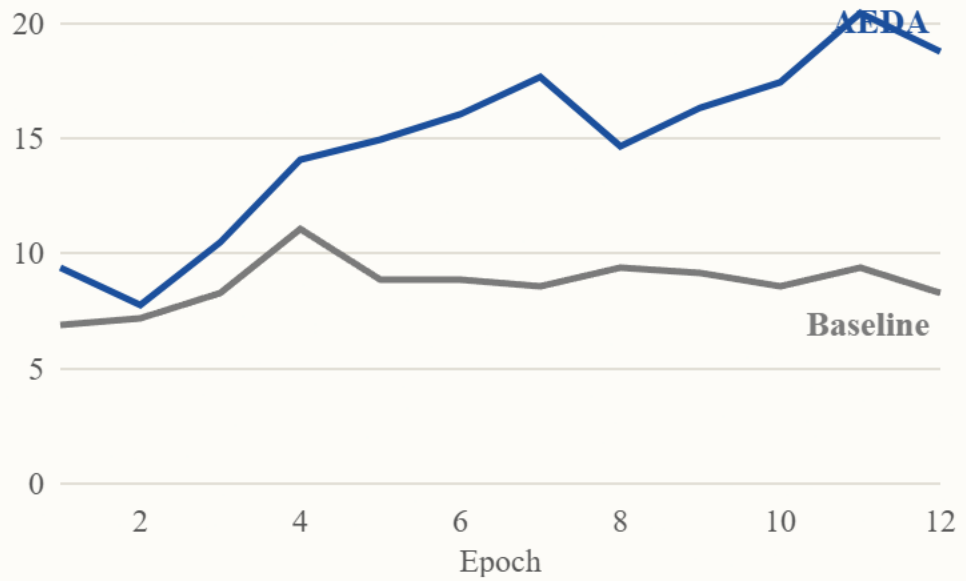
	Baseline	Baseline + AEDA
Training resumes	1,728	5,184 (original + 2 augmented copies each)
Model	2-layer LSTM, 128-d embeddings, 256 hidden units	same
Regularization	none	none
Optimizer	Adam, lr 1e-3, batch 64	same
Epochs / seed	12 / 42	12 / 42
Validation / test	362 / 394	same, not augmented

With AEDA, validation loss climbs much more slowly

Validation loss



Validation accuracy (%)



Test accuracy doubled, but both models are still weak

	Baseline	AEDA
Final train accuracy	91.7%	68.5%
Final validation accuracy	8.3%	18.8%
Train minus validation	83.4 points	49.7 points
Lowest val loss (epoch)	3.127 (2)	3.045 (4)
Val loss rise after lowest	+2.10	+0.56
Test accuracy	10.2%	20.3%
Test macro F1	9.0%	17.7%

Test set: 394 resumes. Chance level with 24 classes is about 4%.

AEDA helped, but it did not solve the problem

What we found

The baseline overfits from epoch 2. With AEDA the lowest validation loss comes at epoch 4 and the rise afterwards is about a quarter as large.

Test accuracy and macro F1 both roughly doubled.

Limits

- One run with one seed.
- AEDA does 3x more updates per epoch.
- Its train accuracy is on augmented data.
- 20% accuracy is still low. Using only the last LSTM state after padding is a likely cause.

Next: more seeds, different numbers of augmented copies, then dropout or pretrained embeddings.